

AIM:

TO write an Assembly language program for a
ASCII to Hexadecimal code conversion by using the
8085 MP.

APPARATUS REQUIRED:

1. Microprocessor kit.
2. power supply.

ALGORITHM:

1. Start the program.
2. Initialize 0000 to the Accumulator & then load the
9000 address to Accumulator content.
3. Subtract the ASCII code 30 from Accumulator value
of 9000.
4. Compare the Accumulator with 0A. If the carry
is set, then jump to loop.
5. Similarly subtract the 04 from Accumulator value
& store the result.



ASSEMBLY LANGUAGE PROGRAM CHART

CODE CONVERSION
ASCII TO HEXADECIMAL

ADDRESS	LABEL	MINEMONICS	OPCODE	COMMENTS
8000		LDA 9000	3A,00,90	Load the content of 9000 to accumulator.
8003		CPI 40 30	FE,40	Compare accumulator with 40.
8005		JC store	DA,0A,80	If carry is set, then jump to store.
8008		SUI 07	D6,07	Subtract 07 from accumulator.
800A	store	SUI 30	D6,30	Subtract 30 from accumulator.
800C		STA 9001	32,01,90	Store the result in 9001.
800F		HLT	76	Stop the program.

$$\frac{5}{2}p$$

0000

1

39

d/p

9001

1

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